Joints and classification

JOINTS AND CLASSIFICATION (sample notes)

Joints are classified structurally and functionally.

Structural classes
- Fibrous joints: bones joined by dense connective tissue; little or no movement. Examples: sutures of the skull, the gomphosis of a tooth in its socket.
- Cartilaginous joints: bones joined by cartilage. Synchondroses (for example, the epiphyseal plate) and symphyses (for example, the pubic symphysis, intervertebral discs).
- Synovial joints: freely movable; contain a joint capsule, synovial fluid and articular cartilage. Examples: knee, shoulder, hip.

Functional classes
- Synarthrosis: immovable. Amphiarthrosis: slightly movable. Diarthrosis: freely movable (all synovial joints).

Synovial joint types by movement
- Hinge (elbow), pivot (atlantoaxial joint), ball-and-socket (hip, shoulder), saddle (thumb carpometacarpal), condyloid (wrist), plane (intercarpal).

Clinical link: the rotator cuff stabilises the shoulder, a ball-and-socket joint with a shallow socket, which makes it highly mobile but prone to dislocation.